
UNIT 8 INTERNET OF THINGS

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8.0 INTRODUCTION

In the earlier blocks, we had studied various concepts on Cloud Computing. In this unit, we will focus on another important technology which is becoming very much popular namely Internet of Things.

Over the past few years, IoT has become one of the most important technologies of the 21st century. The ***Internet of Things***, often abbreviated as ***IoT***, represents a revolutionary concept that has transformed the way we perceive and interact with technology. The Internet of Things (IoT) describes the network of physical objects - “things”- that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet. These devices range from ordinary household objects to sophisticated industrial tools. With more than 7 billion connected IoT devices today, experts are expecting this number to grow to 10 billion by 2020 and 22 billion by 2025.

In its essence, IoT refers to a vast network of interconnected devices, sensors, machines, and systems that communicate and share data with each other over the internet. These devices can range from everyday objects like household appliances, wearable devices, and cars to industrial machinery and infrastructure components.

What distinguishes IoT is its ability to enable these devices to collect and exchange data, allowing for seamless integration, automation, and intelligent decision-making. By embedding sensors, software, and connectivity into various objects, IoT empowers them to gather real-time information, analyze it, and respond intelligently, fostering efficiency, convenience, and innovation across numerous industries and daily life. The potential applications of IoT are extensive, spanning across healthcare, agriculture, transportation, smart cities, manufacturing, and more. Through IoT, devices can autonomously perform tasks, optimize processes, and enhance overall experiences for individuals and businesses alike. However, as IoT continues to evolve and expand, considerations regarding security, privacy, and standardization remain critical. Safeguarding data, ensuring interoperability between devices, and establishing robust protocols are pivotal for the sustainable growth and safe implementation of IoT technologies.

Internet of Things (IoT) is a massive network of physical devices embedded with sensors, software, electronics, and network which allows the devices to exchange or collect data and perform certain actions.

IoT comprises “*Internet*” and “*Things*”, wherein:

- *Things* refer to physical devices, appliances, and gadgets.
- *Internet* denotes the connectivity facilitating the interaction between these devices.

The primary objective of IoT is to extend internet connectivity beyond traditional devices like computers and smartphones, encompassing various devices used in homes and businesses. This technology enables remote control of devices through network infrastructure, reducing human effort and facilitating seamless access to connected devices. By enabling autonomous control, these devices can function without direct human interaction. Leveraging AI algorithms, data collection, and network connectivity, IoT enhances our lives by imbuing these devices with virtual intelligence.

Ultimately, the Internet of Things stands as a pivotal force reshaping our world, promising a future where the integration of smart devices not only augments our capabilities but also transforms the very fabric of how we live, work, and interact with the world around us. In this unit we will focus on IoT and its characteristics, categories of IoT, its enablers and connectivity layers, baseline technologies of IoT and IoT architecture.

8.1 OBJECTIVES

After going through this unit, you shall be able to:

- understand various IoT and its characteristics;
- list and describe various IoT categories;
- describe the baseline technologies of IoT;
- discuss IoT architecture;
- elucidate the applications of IoT; and
- explain the challenges in IoT.

8.2 IoT AND ITS CHARACTERISTICS

The *Internet of Things (IoT)* is a vast network of interconnected physical devices, vehicles, appliances, and other objects embedded with sensors, software, and connectivity, allowing them to collect and exchange data. These devices communicate with each other and with central systems through the internet, enabling them to gather information, make intelligent decisions, and perform tasks more efficiently.

IoT devices range from everyday consumer products like smart thermostats, wearable fitness trackers, and home assistants to complex industrial machinery used in manufacturing, healthcare equipment, and infrastructure monitoring systems. They are designed to sense specific aspects of their environment, gather data, and often act on that data through automated processes.

The power of IoT lies in its ability to connect these devices, analyze the data they generate in real-time, and utilize insights to improve efficiency, productivity, and convenience across various industries and in daily life. For instance, in agriculture, IoT devices can monitor soil moisture levels and weather conditions to optimize irrigation, while in healthcare; wearable devices can track vital signs and provide proactive health monitoring.

IoT technology continues to evolve, offering innovative solutions in areas like smart homes, transportation, agriculture, healthcare, manufacturing, and more. As it expands, considerations about security, privacy, data management, and interoperability are crucial to ensuring the safe and effective integration of IoT devices into our lives and industries.

8.2.1 Characteristics of IoT

Following are the characteristics collectively define the capabilities and functionalities of IoT, contributing to its widespread adoption and application across various industries and everyday life scenarios:

- **Connectivity:** IoT devices are interconnected through the internet, enabling seamless communication and data exchange between devices, systems, and users.
- **Sensing and Monitoring:** IoT devices are equipped with sensors to collect data from their environment, enabling them to monitor and gather real-time information about various conditions.
- **Data Processing:** IoT devices often have the capability to process data locally or transmit it to centralized systems for analysis. They can perform basic to complex data processing tasks depending on their capabilities.
- **Automation and Control:** IoT allows for remote control and automation of devices, enabling actions to be performed without direct human intervention. This feature enhances efficiency and convenience in various applications.
- **Scalability:** IoT systems are designed to scale, accommodating a vast number of devices and users, ensuring seamless integration and operation as the network grows.

- **Interoperability:** Devices from different manufacturers and with various functionalities can communicate and work together, thanks to standardized protocols and interfaces, ensuring compatibility and smooth operation within IoT ecosystems.
- **Security and Privacy:** Ensuring the security of data transmitted and stored by IoT devices is crucial. Security measures such as encryption, authentication, and access control are implemented to protect data and maintain user privacy.
- **Energy Efficiency:** Many IoT devices are designed to operate on low power and are energy-efficient, prolonging their operational life and reducing environmental impact.
- **Dynamic and Self Adapting:** IoT devices and systems may have the capability to dynamically adapt with the changing contexts and take actions based on their operating conditions, user's context or sensed environment. Eg: the surveillance system is adapting itself based on context and changing conditions.
- **Self Configuring:** allowing a large number of devices to work together to provide certain functionality.
- **Unique Identity:** Each IoT device has a unique identity and a unique identifier (IP address).
- **Integrated into Information Network:** that allow them to communicate and exchange data with other devices and systems.

8.2.3 Technologies That Made IoT Possible

Although the concept of IoT has existed for some time, recent advancements across diverse technologies have rendered it feasible.

- Accessibility to cost-effective, energy-efficient sensor technology is broadening the potential for IoT implementation among a wider array of manufacturers.
- Connectivity options, supported by various internet network protocols, have simplified the integration of sensors with cloud systems and other interconnected devices, facilitating swift and efficient data transmission.
- The proliferation of cloud computing platforms has democratized access to scalable infrastructure, enabling businesses and consumers to expand operations without the burden of managing complex systems.
- Progress in machine learning and analytics, coupled with extensive data repositories stored in the cloud, empowers businesses to swiftly glean valuable insights from the burgeoning volumes of IoT-generated data, constantly pushing the boundaries of IoT capabilities.
- The evolution of conversational artificial intelligence (AI), enabled by advancements in neural networks, has brought natural-language processing (NLP) to IoT devices like Alexa, Cortana, and Siri, making these digital assistants both appealing and practical for home use at an affordable rate.

8.3 IoT CATEGORIES

Following categories illustrate the diverse applications of IoT across various industries, each aimed at improving efficiency, productivity, and overall quality of life through the integration of connected devices and smart technologies.

- **Consumer IoT (CIoT)** refers to the use of IoT for consumer applications and devices. Common CIoT products include smartphones, wearables, smart assistants, home appliances, etc. Typically, CIoT solutions leverage Wi-Fi, Bluetooth, and ZigBee to facilitate connectivity. These technologies offer short-range communication suitable for deployments in smaller venues, such as homes and offices.

While CIoT tends to focus on augmenting personal and home environments, Commercial IoT goes a bit further, delivering the benefits of IoT to larger venues. Think: commercial office buildings, supermarkets, stores, hotels, healthcare facilities, and entertainment venues.

There are numerous use cases for commercial IoT, including monitoring environmental conditions, managing access to corporate facilities, and economizing utilities and consumption in hotels and other large venues. Many Commercial IoT solutions are geared towards improving customer experiences and business conditions.

- **Industrial IoT (IIoT)**, is perhaps the most dynamic wing of the IoT industry. Its focus is on augmenting existing industrial systems, making them both more productive and more efficient. IIoT deployments are typically found in large-scale factories and manufacturing plants and are often associated with industries like healthcare, agriculture, automotive, and logistics. The Industrial Internet is perhaps the most well-known example of IIoT.
- **Infrastructure IoT** is concerned with the development of smart infrastructures that incorporate IoT technologies to boost efficiency, cost savings, maintenance, etc. This includes the ability to monitor and control operations of urban and rural infrastructures, such as bridges, railway tracks, and on- and offshore windfarms. Technically speaking, infrastructure IoT is a subset of IIoT. However, due to its significance, it's often treated as its own separate thing.
- **Internet of Military Things (IoMT)**, often referred to as Battlefield IoT, the Internet of Battlefield Things, or simply IoBT. IoMT is precisely what it sounds like — the use of IoT in military settings and battlefield situations. It is chiefly aimed at increasing situational awareness, bolstering risk assessment, and improving response times. Common IoMT applications include connecting ships, planes, tanks, soldiers, drones, and even Forward Operating Bases via an interconnected system. In addition, IoMT produces data that can be

leveraged to improve military practices, systems, equipment, and strategy.

- **Retail IoT (RIoT)**, is utilized for inventory management, smart shelves, personalized shopping experiences, and customer behavior analytics. It enhances operational efficiency and improves the customer shopping journey.
- **Environmental IoT**, involves the use of IoT devices to monitor and manage environmental factors such as air quality, water quality, and climate conditions. Sensors collect data to enable better environmental management and sustainability efforts.

8.4 IoT ENABLERS AND CONNECTIVITY LAYERS

Installers, repair experts, artisans, electrical technicians, plumbers, and architects are all involved in linking devices and systems to the internet, whether for personal, commercial, or other business purposes.

The Internet of Things (IoT) empowers devices to make informed decisions that drive favorable business outcomes, and at the heart of these decisions are sensors. As the pressures of cost and time-to-market escalate, sensors offer enhanced visibility into interconnected systems, enabling these systems to intelligently respond to changes originating from both external forces and internal factors.

Sensors play a pivotal role in furnishing actionable insights that fuel the IoT, empowering organizations to execute more astute business decisions. Real-time measurements facilitated by sensors serve as the catalyst for the transformative potential of the IoT in an organization's adaptability to change.

Wi-Fi initially catered to computers, while 4G LTE wireless was tailored for smartphones and portable devices, both achieving tremendous success by aligning with their intended devices. Similarly, 5G has been designed as the inaugural wireless technology with the particular focus on extremely small, low-power, and nearly omnipresent IoT devices.

Unlike Wi-Fi and LTE devices, which are handled and connected to power sources daily, IoT sensors operate autonomously for extended periods, often in remote and inaccessible locations, without requiring recharging or replacement. The advent of the IoT is driving the development of numerous 5G communication standards, spawning a multitude of network types rather than a singular or dual protocol.

☞ Check Your Progress 1

1) What is the Internet of Things (IoT)?

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2) What is the purpose of IoT?

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8.5 BASELINE TECHNOLOGIES OF IoT

The foundation of the Internet of Things (IoT) is built upon several key technologies that enable the connectivity, data processing, and communication between devices. Some of the base technologies of IoT include:

- **Sensors and Actuators:** These are fundamental components that gather data from the physical world (sensors) and perform actions based on received instructions (actuators). Sensors collect information such as temperature, motion, light, and more, while actuators execute commands like opening a valve or turning on a motor.
- **Connectivity:** Various communication protocols and technologies facilitate the transfer of data between IoT devices and systems. This includes Wi-Fi, Bluetooth, Zigbee, Z-Wave, RFID, NFC, cellular networks (2G, 3G, 4G, and now 5G), LoRaWAN, and others. Each protocol has its strengths and is suited to different IoT use cases based on factors like range, power consumption, and bandwidth.
- **Cloud Computing:** Cloud platforms store and process the vast amounts of data generated by IoT devices. These platforms provide the infrastructure for data storage, analysis, and enable remote access to information from anywhere. Cloud services also offer scalability and computational power necessary for handling IoT-generated data.
- **Edge Computing:** To reduce latency and handle data processing closer to the source, edge computing is utilized. It involves processing data locally on devices or within the local network instead of sending all information to the cloud. This enhances real-time decision-making and reduces the load on central servers.
- **Security Technologies:** IoT devices require robust security measures to protect data and prevent unauthorized access. Encryption protocols,

authentication methods, firewalls, and secure device management frameworks are crucial components in securing IoT ecosystems.

- **Machine Learning and AI:** Advanced analytics, machine learning algorithms, and artificial intelligence play a significant role in deriving insights from the massive volumes of IoT-generated data. These technologies enable predictive maintenance, pattern recognition, anomaly detection, and intelligent decision-making within IoT systems.
- **IoT Analytics:** IoT business models will exploit the information collected by “things” in many ways, which will demand new analytic tools and algorithms. As data volumes increase over the next five years, the needs of the IoT may diverge further from traditional analytics.
- **IoT Device (Thing) Management:** Long-lived nontrivial “things” will require management and monitoring, including device monitoring, firmware and software updates, diagnostics, crash analysis and reporting, physical management, and security management. Tools must be capable of managing and monitoring thousands and perhaps even millions of devices.
- **Low-Power, Short-Range IoT Networks:** Low-power, short-range networks will dominate wireless IoT connectivity through 2025, far outnumbering connections using wide-area IoT networks. However, commercial and technical trade-offs mean that many solutions will coexist, with no single dominant winner.
- **IoT Processors:** The processors and architectures used by IoT devices define many of their capabilities, such as whether they are capable of strong security and encryption, power consumption, whether they are sophisticated enough to support an operating system, updatable firmware, and embedded device management agents. Understanding the implications of processor choices will demand deep technical skills.
- **IoT Operating Systems:** Traditional operating systems such as Windows and iOS were not designed for IoT applications. They consume too much power, need fast processors, and in some cases, lack features such as guaranteed real-time response. They also have too large a memory footprint for small devices and may not support the chips that IoT developers use. Consequently, a wide range of IoT-specific operating systems has been developed to suit many different hardware footprints and feature needs.
- **Event Stream Processing:** Some IoT applications will generate extremely high data rates that must be analyzed in real time. Systems creating tens of thousands of events per second are common, and millions of events per second can occur in some situations. To address such requirements, distributed stream computing platforms have

emerged that can process very high-rate data streams and perform tasks such as real-time analytics and pattern identification.

- **IoT Standards and Ecosystems:** Standards and their associated application programming interfaces (APIs) will be essential because IoT devices will need to interoperate and communicate, and many IoT business models will rely on sharing data between multiple devices and organizations. Many IoT ecosystems will emerge, and organizations creating products may have to develop variants to support multiple standards or ecosystems and be prepared to update products during their life span as the standards evolve and new standards and APIs emerge.

8.6 SENSORS

Sensors are used for sensing things and devices etc. A sensor is a device that provides a usable output in response to a specified measurement. The sensor attains a physical parameter and converts it into a signal suitable for processing (e.g. electrical, mechanical, optical) the characteristics of any device or material to detect the presence of a particular physical quantity. The output of the sensor is a signal which is converted to a human-readable form like changes in characteristics, changes in resistance, capacitance, impedance etc.

8.6.1. Characteristics of a Sensor

The static accuracy of a sensor indicates how much the sensor signal correctly represents the measured quantity after it stabilizes (i.e. beyond the transient period.) Important **static characteristics** of sensors include sensitivity, resolution, linearity, zero drift and full-scale drift, range, repeatability and reproducibility.

1. *Sensitivity* is a measure of the change in output of the sensor relative to a unit change in the input (the measured quantity.) Example: The speakers you purchase for your home entertainment may have a rated sensitivity of 89 dB Signal Pressure Level per Watt per meter.
2. *Resolution* is the smallest amount of change in the input that can be detected and accurately indicated by the sensor. Example: What is the resolution of an ordinary ruler? of a Vernier Calipers?
3. *Linearity* is determined by the calibration curve. The static calibration curve plots the output amplitude versus the input amplitude under static conditions. Its degree of resemblance to a straight line describes the linearity.
4. *Drift* is the deviation from a specific reading of the sensor when the sensor is kept at that value for a prolonged period of time. The *zero drift* refers to the change in sensor output if the input is kept steady at a level that (initially) yields a zero reading. Similarly, the *full -scale drift* is the drift if the input is maintained at a value which originally

yields a full scale deflection. Reasons for drift may be extraneous, such as changes in ambient pressure, humidity, temperature etc., or due to changes in the constituents of the sensor itself, such as aging, wear etc.

5. The *range* of a sensor is determined by the allowed lower and upper limits of its input or output. Usually the range is determined by the accuracy required. Example: Sometimes the range may just be determined by physical limitations. Example: a pocket ruler.
6. *Repeatability* is defined as the deviation between measurements in a sequence when the object under test is the same and approaches its value from the same direction each time. The measurements have to be made under a short enough time duration so as not to allow significant long term drift. Repeatability is usually specified as a percentage of the sensor range. Example:
7. *Reproducibility* is the same as repeatability, except it also incorporates long time lapses between subsequent measurements. The sensor has to be operation between measurements, but must be calibrated. Reproducibility is specified as a percentage of the sensor range per unit of time.

The **dynamic characteristics** of a sensor represent the time response of the sensor system. Knowledge of these is essential to fruitfully use a sensor. Important common dynamic responses of sensors include rise time, delay time, peak time, settling time percentage error and steady-state error

8.6.2. Classification of Sensors

The common IoT sensors include:

Temperature sensors, Pressure sensors, Motion sensors, Level sensors, Image sensors, Proximity sensors, Water quality sensors, Chemical sensors, Gas sensors, Smoke sensors, Infrared (IR) sensors, Humidity sensors, etc.

A description of each of these sensors is provided below.

Temperature sensors

Temperature sensors detect the temperature of the air or a physical object and convert that temperature level into an electrical signal that can be calibrated accurately reflect the measured temperature. These sensors could monitor the temperature of the soil to help with agricultural output or the temperature of a bearing operating in a critical piece of equipment to sense when it might be overheating or nearing the point of failure.

Pressure sensors

Pressure sensors measure the pressure or force per unit area applied to the sensor and can detect things such as atmospheric pressure, the pressure of a stored gas or liquid in a sealed system such as tank or pressure vessel, or the **weight of an object**.

Motion sensors

Motion sensors or detectors can sense the movement of a physical object by using any one of several technologies, including passive infrared (PIR), microwave detection, or ultrasonic, which uses sound to detect objects. These sensors can be used in security and intrusion detection systems, but can also be used to automate the control of doors, sinks, air conditioning and heating, or other systems.

Level sensors

Level sensors translate the level of a liquid relative to a benchmark normal value into a signal. Fuel gauges display the level of fuel in a vehicle's tank, as an example, which provides a continuous level reading. There are also point level sensors, which are a go-no/go or digital representation of the level of the liquid. Some automobiles have a light that illuminates when the fuel level tank is very close to empty, acting as an alarm that warns the driver that fuel is about to run out completely.

Image sensors

Image sensors function to capture images to be digitally stored for processing. License plate readers are an example, as well as facial recognition systems. Automated production lines can use image sensors to detect issues with quality such as how well a surface is painted after leaving the spray booth.

Proximity sensors

Proximity sensors can detect the presence or absence of objects that approach the sensor through a variety of different technology designs.

Water quality sensors

The importance of water to human beings on earth not only for drinking but as a key ingredient needed in many production processes dictates the need to be able to sense and measure parameters around water quality. Some examples of what is sensed and monitored include:

Chemical presence (such as chlorine levels or fluoride levels), Oxygen levels (which may impact the growth of algae and bacteria), Electrical conductivity (which can indicate the level of ions present in water), pH level (a reflection of the relative acidity or alkalinity of the water), Turbidity levels (a measurement of the amount of suspended solids in water)

Chemical sensors

Chemical sensors are designed to detect the presence of specific chemical substances which may have inadvertently leaked from their containers into spaces that are occupied by personnel and are useful in controlling industrial process conditions.

Gas sensors

Related to chemical sensors, gas sensors are tuned to detect the presence of combustible, toxic, or flammable gas in the vicinity of the sensor. Examples of specific gases that can be detected include:

Bromine (Br₂), Carbon Monoxide (CO), Chlorine (Cl₂), Chlorine Dioxide (ClO₂), Hydrogen Cyanide (HCN), Hydrogen Peroxide (H₂O₂), Hydrogen Sulfide (H₂S), Nitric Oxide (NO), Nitrogen Dioxide (NO₂), Ozone (O₃), etc.

Smoke sensors

Smoke sensors or detectors pick up the presence of smoke conditions which could be an indication of a fire typically using optical sensors (photoelectric detection) or ionization detection.

Infrared (IR) sensors

Infrared sensor technologies detect infrared radiation that is emitted by objects. Non-contact thermometers make use of these types of sensors as a way of measuring the temperature of an object without having to directly place a probe or sensor on that object. They find use in analyzing the heat signature of electronics and detecting blood flow or blood pressure in patients.

Acceleration sensors

While motion sensors detect movement of an object, acceleration sensors, or accelerometers as they are also known, detect the rate of change of velocity of an object. This change may be due to a free-fall condition, a sudden vibration that is causing movement with speed changes, or rotational motion (a directional change).

8.7 ACTUATORS

An actuator is a machine component or system that moves or controls the mechanism or the system. Sensors in the device sense the environment, then control signals are generated for the actuators according to the actions needed to perform. Actuators convert an electrical signal into a corresponding physical quantity such as movement, force, sound etc.

8.7.1. Types of Actuators

Actuators are devices used in various engineering and technological applications to convert energy into physical movement or action. There are several types of actuators, each designed for specific purposes and with different mechanisms for generating motion. Here are some common types:

1. Electric Actuators

- **Linear Actuators:** These actuators produce linear motion, converting rotary motion from an electric motor into linear movement. They are used in applications like opening and closing doors, adjusting solar panels, or controlling valves.
- **Rotary Actuators:** Convert electrical energy into rotational motion. They're used in applications such as robotics, conveyor systems, and control mechanisms for valves or dampers.



Figure 1: Electric Actuator

2. Hydraulic Actuators

- **Linear Hydraulic Actuators:** Utilize pressurized hydraulic fluid to generate linear motion. They're employed in heavy machinery, construction equipment, and industrial systems that require high force output.
- **Rotary Hydraulic Actuators:** Convert hydraulic power into rotary motion. They're used in heavy-duty applications like moving heavy loads, steering systems, and some types of machinery.

3. Pneumatic Actuators

- **Pneumatic Cylinders:** Use compressed air to create linear motion. They're commonly found in industrial automation, such as in manufacturing, automotive assembly lines, and control systems.
- **Pneumatic Rotary Actuators:** Generate rotary motion using compressed air. They're used in applications where quick and precise rotation is required, such as in robotics and manufacturing.



Figure 2: Pneumatic Actuator

4. Piezoelectric Actuators

- These actuators use piezoelectric materials that change shape when an electric field is applied. They are used in precision positioning systems, nanopositioning, and in some medical devices.

5. Electroactive Polymers (EAPs)

- EAP actuators change shape or size in response to electrical stimulation. They're still in the development stage but hold promise for artificial muscles, soft robotics, and biomimetic devices.

Actuators play a crucial role in a wide range of (IoT) applications, from manufacturing and robotics to aerospace, automotive, and healthcare. The choice of actuator depends on the specific requirements of the application, including the type of motion needed, force or torque requirements, precision, speed, and environmental considerations.

8.7.2 Applications of Actuators

Actuators have a wide range of use in the modern world in machines, automobiles, and automation. The following table 1 describes common applications, devices suitable for said applications, and actuators that provide power to the devices.

Table 1: Applications, Devices and Types of Actuators

Application	Device	Actuator Type
Automated control of fluid flow in pipelines and process systems	Control Valve, Flow Meter	Linear, Rotary (Hydraulic, Electric)
Adjustment of industrial valves, positioning of machine components	Ball Valve, Solenoid Valve, Servo Motor	Rotary (Hydraulic, Electric)
Digging, grading, and excavating in construction and mining operations	Excavator, Backhoe	Linear, Rotary (Hydraulic)
Manufacturing of metal parts, plastic molding, and forging operations	Hydraulic Press, CNC Machine, Forging Hammer	Rotary (Hydraulic, Electric)
Powering machine tools, robots,	Electric Motor, Robot Arm,	Linear, Rotary

and conveyor systems	Conveyor Belt	(Electric, Hydraulic)
Positioning of machine components in automated production systems	Linear Actuator, Servo Motor, Gripper	Linear, Rotary (Electric, Hydraulic)
Regulating the flow of fuel and air into internal combustion engines	Throttle Valve, Fuel Injector	Rotary (Mechanical, Electric)
Regulating the speed of steam or gas turbines in power plants	Turbine Governor, Valve	Rotary (Electric, Hydraulic, Thermal)
Simple machine control in mechanical systems, such as door openers	Mechanical Lever, Electric Switch	Linear, Rotary (Mechanical, Electric)
Transmission of power in machines, such as conveyor systems and gear pumps	Gearbox, Gear Pump, Hydraulic Motor	Linear, Rotary (Hydraulic, Mechanical, Electric)

8.8 ARDUINO BOARD AND RASPBERRY PI

Arduino board and Raspberry Pi are both popular hardware platforms used in electronics and computing. Arduino is more focused on hardware interaction and real-time processing for embedded systems, Raspberry Pi serves as a miniature computer suitable for a wide array of applications, from education to IoT and multimedia projects, owing to its computational capabilities and versatility.

Arduino Board

- An Arduino is actually a microcontroller based kit.
- It is basically used in communications and in controlling or operating many devices.
- Arduino UNO board is the most popular board in the Arduino board family.
- In addition, it is the best board to get started with electronics and coding.
- Some boards look a bit different from the one given below, but most Arduino's have majority of these components in common.
- It consists of two memories- Program memory and the data memory.
- The code is stored in the flash program memory, whereas the data is stored in the data memory.
- Arduino Uno consists of 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header, and a reset button
1.Power USB 2.Power (Barrel Jack) 3.Voltage Regulator 4.Crystal Oscillator 17.Arduino Reset 5.Arduino Reset 6,7,8,9.Pins (3.3, 5, GND, Vin) 10.Analog pins 11.Main microcontroller 12.ICSP pin 13.Power LED indicator 14.TX and RX LEDs 15.D.

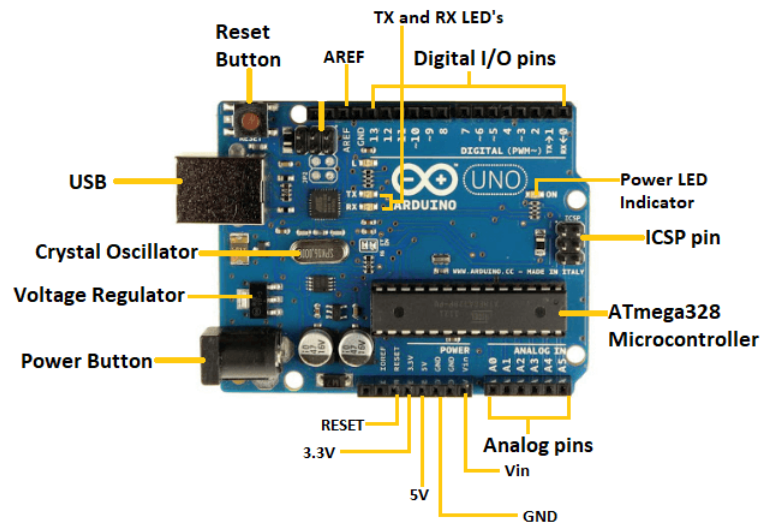


Figure 3: Arduino Board

Raspberry Pi

- The Raspberry Pi is a very cheap computer that runs Linux, but it also provides a set of GPIO (general purpose input/output) pins that allow you to control electronic components for physical computing and explore the Internet of Things (IoT).
- Raspberry Pi was basically introduced in 2006.
- It is particularly designed for educational use and intended for Python.
- A Raspberry Pi is of small size i.e., of a credit card sized single board computer, which is developed in the United Kingdom(U.K) by a foundation called Raspberry Pi

There have been three generations of Raspberry Pis: Pi 1, Pi 2, and Pi 3

- The first generation of Raspberry (Pi 1) was released in the year 2012, that has two types of models namely model A and model B.
- Raspberry Pi can be plugged into a TV, computer monitor, and it uses a standard keyboard and mouse.
- It is user friendly as can be handled by all the age groups.
- It does everything you would expect a desktop computer to do like word-processing, browsing the internet spread sheets, playing games to playing high definition videos.
- All models feature on a broadcom system on a chip (SOC), which includes chip graphics processing unit GPU(a Video Core IV), an ARM compatible and CPU.
- The CPU speed ranges from 700 MHz to 1.2 GHz for the Pi 3 and on board memory range from 256 MB to 1 GB RAM.
- An operating system is stored in the secured digital SD cards and program memory in either the MicroSDHC or SDHC sizes.
- Most boards have one to four USB slots, composite video output, HDMI and a 3.5 mm phone jack for audio. Some models have WiFi and Bluetooth.

- All models feature a Broadcom system on a chip (SoC) with an integrated ARM-compatible central processing unit (CPU) and on-chip graphics processing unit (GPU).
- Processor speed ranges from 700 MHz to 1.4 GHz for the Pi 3 Model B+ or 1.5 GHz for the Pi 4; on-board memory ranges from 256 MB to 1 GB with up to 4 GB available on the Pi 4 random-access memory (RAM).
- Secure Digital (SD) cards in Micro SDHC form factor (SDHC on early models) are used to store the operating system and program memory.
- The boards have one to five USB ports. For video output, HDMI and composite video are supported, with a standard 3.5 mm tip-ring-sleeve jack for audio output.
- Lower-level output is provided by a number of GPIO pins, which support common protocols like I²C. The B-models have an 8P8C Ethernet port and the Pi 3 and Pi Zero W have on-board Wi-Fi and Bluetooth.

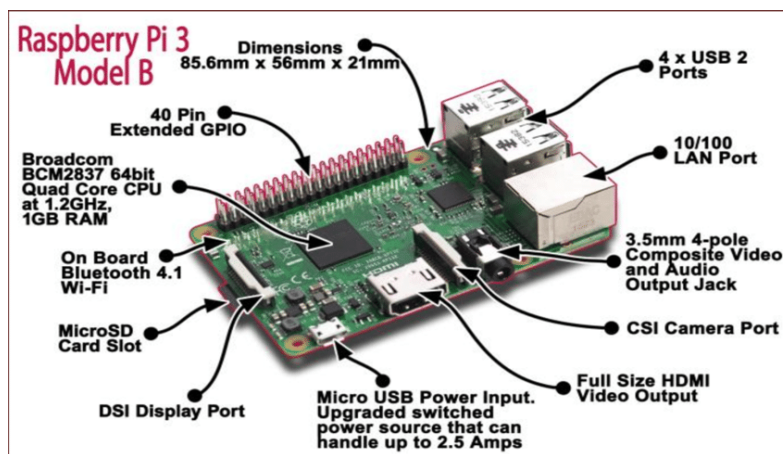


Figure 4: Raspberry Pi

☞ Check Your Progress 2

1) What are the components of an IoT system?

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2) How do IoT devices communicate?

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3) List the differences of Arduino Board and Raspberrypi in a table.

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8.9 IoT ARCHITECTURE

The Reference Model introduced in 2014 by Cisco, IBM, and Intel at the 2014 IoT World Forum has as many as seven layers. According to an official [press release](#) by Cisco forum host, the architecture aims to “*help educate CIOs, IT departments, and developers on deployment of IoT projects, and accelerate the adoption of IoT.*” These layers are:

- The **perception layer** hosting smart things;
- The **connectivity or transport layer** transferring data from the physical layer to the cloud and vice versa via networks and gateways;
- The **processing layer** employing IoT platforms to accumulate and manage all data streams; and
- The **application layer** delivering solutions like analytics, reporting, and device control to end users.

(i) Perception layer: Converting analog signals into digital data and vice versa

The initial stage of any IoT system embraces a wide range of “things” or endpoint devices that act as a bridge between the real and digital worlds. They vary in form and size, from tiny silicon chips to large vehicles. By their functions, IoT things can be divided into the following large groups.

- **Sensors** such as probes, gauges, meters, and others. They collect physical parameters like temperature or humidity, turn them into electrical signals, and send them to the IoT system. IoT sensors are typically small and consume little power.
- **Actuators**, translating electrical signals from the IoT system into physical actions.
- **Machines and devices** connected to sensors and actuators or having them as integral parts.

(ii) Connectivity layer: enabling data transmission

The second level is in charge of all communications across devices, networks, and cloud services that make up the IoT infrastructure. The connectivity between the physical layer and the cloud is achieved in two ways:

- directly, using TCP or UDP/IP stack;
- via gateways — hardware or software modules performing translation between different protocols as well as encryption and decryption of IoT data.

The communications between devices and cloud services or gateways involve different networking technologies.

Ethernet connects stationary or fixed IoT devices like security and video cameras, permanently installed industrial equipment, and gaming consoles.

WiFi, the most popular technology of wireless networking, is a great fit for data-intensive IoT solutions that are easy to recharge and operate within a

small area. A good example of use is smart home devices connected to the electrical grid.

NFC (Near Field Communication) enables simple and safe data sharing between two devices over a distance of 4 inches (10 cm) or less.

Bluetooth is widely used by wearables for short-range communications. To meet the needs of low-power IoT devices, the Bluetooth Low-Energy (BLE) standard was designed. It transfers only small portions of data and doesn't work for large files.

LPWAN (Low-power Wide-area Network) was created specifically for IoT devices. It provides long-range wireless connectivity on low power consumption with a battery life of 10+ years. Sending data periodically in small portions, the technology meets the requirements of smart cities, smart buildings, and smart agriculture (field monitoring).

ZigBee is a low-power wireless network for carrying small data packages over short distances. The outstanding thing about ZigBee is that it can handle up to 65,000 nodes. Created specifically for home automation, it also works for low-power devices in industrial, scientific, and medical sites.

Cellular networks offer reliable data transfer and nearly global coverage. There are two cellular standards developed specifically for IoT things. LTE-M (Long Term Evolution for Machines) enables devices to communicate directly with the cloud and exchange high volumes of data. NB-IoT or Narrowband IoT uses low-frequency channels to send small data packages.

Edge or Fog computing layer: reducing system latency

This level is essential for enabling IoT systems to meet the speed, security, and scale requirements of the 5th generation mobile network or 5G. The new wireless standard promises faster speeds, lower latency, and the ability to handle many more connected devices, than the current 4G standard.

The idea behind edge or fog computing is to process and store information as early and as close to its sources as possible. This approach allows for analyzing and transforming high volumes of real-time data locally, at the edge of the networks. Thus, you save the time and other resources that otherwise would be needed to send all data to cloud services. The result is reduced system latency that leads to real-time responses and enhanced performance.

Processing layer: making raw data useful

The processing layer accumulates, stores, and processes data that comes from the previous layer. All these tasks are commonly handled via IoT platforms and include two major stages.

Data accumulation stage

The real-time data is captured via an API and put at rest to meet the requirements of non-real-time applications. The data accumulation component stage works as a transit hub between event-based data generation and query-based data consumption.

Among other things, the stage defines whether data is relevant to the business requirements and where it should be placed. It saves data to a wide range of

storage solutions, from data lakes capable of holding unstructured data like images and video streams to event stores and telemetry databases. The total goal is to sort out a large amount of diverse data and store it in the most efficient way.

Data abstraction stage

Here, data preparation is finalized so that consumer applications can use it to generate insights. The entire process involves the following steps: combining data from different sources, both IoT and non-IoT, including ERM, ERP, and CRM systems; reconciling multiple data formats; and aggregating data in one place or making it accessible regardless of location through data virtualization.

Similarly, data collected at the application layer is reformatted here for sending to the physical level so that devices can “understand” it.

Together, the data accumulation and abstraction stages veil details of the hardware, enhancing the interoperability of smart devices. What’s more, they let software developers focus on solving particular business tasks — rather than on delving into the specifications of devices from different vendors.

(iii) Application layer: addressing business requirements

At this layer, information is analyzed by software to give answers to key business questions. There are hundreds of IoT applications that vary in complexity and function, using different technology stacks and operating systems. Some examples are:

device monitoring and control software, mobile apps for simple interactions, business intelligence services, and analytic solutions using machine learning.

Currently, applications can be built right on top of IoT platforms that offer software development infrastructure with ready-to-use instruments for data mining, advanced analytics, and data visualization. Otherwise, IoT applications use APIs to integrate with middleware.

Business layer: Implementing data-driven solutions

The information generated at the previous layers brings value if only it results in problem-solving solution and achieving business goals. New data must initiate collaboration between stakeholders who in turn introduce new processes to enhance productivity.

The decision-making usually involves more than one person working with more than one software solution. For this reason, the business layer is defined as a separate stage, higher than a single application layer.

Security layer: preventing data breaches

It goes without saying that there should be a security layer covering all the above-mentioned layers. IoT security is a broad topic worthy of a separate article. Here we’ll only point out the basic features of the safe architecture across different levels.

Device security: Modern manufacturers of IoT devices typically integrate security features both in the hardware and firmware installed on it. This includes embedded TPM (Trusted Platform Module) chips with cryptographic keys for authentication and protection of endpoint devices;

a secure boot process that prevents unauthorized code from running on a powered-up device; updating security patches on a regular basis; and physical protection like metal shields to block physical access to the device.

Connection security: Whether data is being sent over devices, networks, or applications, it should be encrypted. Otherwise, sensitive information can be read by anybody who intercepts information in transit. IoT-centric messaging protocols like MQTT, AMQP, and DDS may use standard Transport Layer Security (TLS) cryptographic protocol to ensure end-to-end data protection.

Cloud security: Data at rest stored in the cloud must be encrypted as well to mitigate risks of exposing sensitive information to intruders. Cloud security also involves authentication and authorization mechanisms to limit access to the IoT applications. Another important security method is device identity management to verify the device's credibility before allowing it to connect to the cloud.

The good news is that IoT solutions from large providers like Microsoft, AWS, or Cisco come with pre-built protection measures including end-to-end data encryption, device authentication, and access control. However, it always pays to ensure that security is tight at all levels, from the tiniest devices to complex analytical systems.

8.10 APPLICATIONS OF IoT

The Internet of Things (IoT) has an extensive range of applications across various industries. These applications showcase how IoT technology is transforming various aspects of our lives, driving efficiency, automation, and connectivity across different sectors.

- **IoT Wearables**

Wearable technology is a hallmark of IoT applications and probably is one of the earliest industries to have deployed the IoT at its service. We happen to see Fit Bits, heart rate monitors and smart watches everywhere these days.

One of the lesser-known wearables includes the Guardian glucose monitoring device. The device is developed to aid people suffering from diabetes. It detects glucose levels in the body, using a tiny electrode called glucose sensor placed under the skin and relays the information via Radio Frequency to a monitoring device.

- **Smart Home Applications**

When we talk about IoT Applications, Smart Homes are probably the first thing that we think of. The best example I can think of here is *Jarvis*, the AI home automation employed by Mark Zuckerberg. There is also Allen Pan's Home Automation System where functions in the house are actuated by use of a string of musical notes.

- **IoT Applications – Health Care**

IoT applications can turn reactive medical-based systems into proactive wellness-based systems. The resources that current medical research uses, lack critical real-world information. It mostly uses leftover data, controlled environments, and volunteers for medical examination. IoT opens ways to a sea of valuable data through analysis, real-time field data, and testing. The Internet of Things also improves the current devices in power, precision, and availability. IoT focuses on creating systems rather than just equipment.

- **Smart Cities**

By now I assume, most of you must have heard about the term **Smart City**. The hypothesis of the optimized traffic system I mentioned earlier, is one of the many aspects that constitute a smart city.

The thing about the smart city concept is that it's very specific to a city. The problems faced in Mumbai are very different than those in Delhi. The problems in Hong Kong are different from New York. Even global issues, like finite clean drinking water, deteriorating air quality and increasing urban density, occur in different intensities across cities. Hence, they affect each city differently.

- **Agriculture**

Statistics estimate the ever-growing world population to reach nearly 10 billion by the year 2050. To feed such a massive population one needs to marry agriculture to technology and obtain best results. There are numerous possibilities in this field. One of them is the **Smart Greenhouse**.

A greenhouse farming technique enhances the yield of crops by controlling environmental parameters. However, manual handling results in production loss, energy loss, and labor cost, making the process less effective.

- **Industrial Automation**

This is one of the fields where both faster developments, as well as the quality of products, are the critical factors for a higher Return on Investment. With IoT Applications, one could even re-engineer products and their packaging to deliver better performance in both cost and customer experience. IoT here can prove to be game changing with solutions for all the following domains in its arsenal.

- Factory Digitalization
- Product flow Monitoring
- Inventory Management
- Safety and Security
- Quality Control
- Packaging optimization
- Logistics and Supply Chain Optimization

More details you will study in Unit-11 of this course.

8.11 CHALLENGES OF IoT

The biggest challenges for IoT adoption include:

Security Challenges

Rapid advances in both technology and the complexity of cyber-attacks have meant that the risk of security breaches has never been higher. There is an increased responsibility for software developers to create the most secure applications possible to defend against this threat as IoT devices are often seen as easy targets by hackers.

Regulation Challenges

We've already touched on how GDPR has impacted the IoT industry, however, as the industry is still relatively new and young, it generally lacks specific regulation and oversight, which is required to ensure that all devices are produced with a suitable level of protection and security.

Compatibility Challenges

At the core of the IoT concept, all devices must be able to connect and communicate with each other for data to be transferred.

The IoT industry currently lacks any compatibility standards, meaning that many devices could all run on different standards resulting in difficulties communicating with one another effectively.

Bandwidth Challenges

Perhaps at no surprise, devices and applications that rely on the ability to communicate with each other constantly to work effectively tend to use a lot of data at once, leading to bandwidth constraints for those using many devices at once.

Combine this with existing demands for data and broadband in the typical house, and you can quickly see how data and bandwidth limitations can be a challenge.

Customer Expectation Challenges

Arguably the biggest hurdle for the industry relates to customer perception. For anything new to be adopted by the masses, it has to be trusted completely.

For the IoT industry, this is a continuously evolving challenge as it relies on the ability to actively combat security threats and reassure the general consumer market that the devices are both safe to use and secure to hold vast quantities of sensitive data.

Check Your Progress 3

1) What is the architecture of an IoT system?

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2) What is the Perception Layer in IoT?

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3) Explain the Network Layer in IoT.

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4) What is the Application Layer in IoT?

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5) How is data handled in IoT systems?

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6) What are examples of IoT applications?

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8.12 SUMMARY

In this unit we had studied Internet of Things, a new technology.

The Internet of Things (IoT) is a vast network of interconnected devices and sensors that communicate and share data over the internet. Its fundamental concept revolves around enabling everyday objects to collect and exchange information, creating a seamless web of connectivity.

IoT empowers devices to gather real-time data from their surroundings through sensors, process that data, and use it to trigger actions or provide valuable insights. This technology spans across numerous industries, revolutionizing how we live and work.

Key components of IoT include sensors/devices that gather data, connectivity options like Wi-Fi or Bluetooth, data processing and storage systems, and interfaces for user interaction. IoT architecture typically consists of the Perception Layer (sensors), Network Layer (connectivity), and Application Layer (data processing and user interface).

Its applications are diverse, from enhancing daily life through smart homes and wearable tech to optimizing industries via industrial automation, healthcare monitoring, smart cities, and agricultural advancements. Ultimately, IoT aims to drive efficiency, improve decision-making, and create more interconnected and intelligent systems for a wide range of purposes.

8.13 SOLUTIONS / ANSWERS

Check Your Progress 1

1. The Internet of Things refers to a network of interconnected devices that can communicate and share data with each other over the internet.
2. IoT aims to enhance connectivity and automation, allowing devices to collect, exchange, and utilize data to improve efficiency, convenience, and decision-making.

Check Your Progress 2

1. An IoT system typically consists of devices/sensors, connectivity, data processing/storage, and user interface.
2. A: IoT devices communicate using various protocols like Wi-Fi, Bluetooth, Zigbee, or cellular networks to transmit data to other devices or a central hub.
3. The differences define their respective strengths and best use cases. Arduino is excellent for hardware-focused projects and applications requiring real-time interaction, while Raspberry Pi is better suited for more complex computing tasks, running multiple applications, and serving as a miniature computer for various projects. Differences are summarized in the following table:

Parameter	Arduino Board	Raspberry Pi
Purpose	Microcontroller platform for hardware projects	Single-board computer for diverse computing tasks
Processing Power	Generally low-power, suitable for simple tasks	Higher processing power comparable to a computer
Operating	No operating system, runs	Uses various operating systems like

Parameter	Arduino Board	Raspberry Pi
System	Arduino sketches	Linux
Connectivity	Limited connectivity options (USB, Bluetooth)	Extensive connectivity (Wi-Fi, Bluetooth, Ethernet)
GPIO Pins	Abundance of GPIO (General Purpose Input/Output) pins for hardware interaction	GPIO pins available for hardware interfacing
Programming	Uses Arduino IDE and C/C++ for programming	Supports multiple programming languages and IDEs
Price	Generally lower cost	Slightly higher cost, but offers more features
Applications	Ideal for embedded systems, robotics, simple IoT projects	Suitable for media centers, servers, complex IoT applications, and learning programming and computing

Check Your Progress 3

1. IoT architecture comprises three layers: the Perception Layer (sensors/devices), the Network Layer (connectivity), and the Application Layer (data processing and user interface).
2. The Perception Layer includes sensors and devices that collect data from the physical world, such as temperature sensors, cameras, or motion detectors.
3. The Network Layer involves the connectivity infrastructure that enables devices to transmit data. This includes protocols, gateways, and communication technologies.
4. The Application Layer involves data processing, analysis, and the interface that allows users to interact with the system, often through applications or dashboards.
5. Data collected by IoT devices is processed, analyzed, and stored either locally or in the cloud, where it can be accessed and utilized for various applications.
6. Smart homes, wearable devices, industrial automation, smart cities, healthcare monitoring, and agriculture are examples of IoT applications impacting various sectors.

(More on IoT applications and application development can be studied in Unit-12).

8.14 FURTHER READINGS

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